



Effects of congestion surcharges: From ridership to competition and safety

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ABSTRACT

This paper examines the effects of a 2019 congestion surcharge on for-hire-vehicle and taxi usage in New York City. We use a difference-in-differences method to evaluate both the change in rides during this period and the coinciding decline in pickups. We find a significant decline in rides originating from the charged area (11%), and a parallel reduction in collisions (5%), injuries (9%), and some fatalities, suggesting that the policy has successfully created the intended reduction in vehicular activity. Taxis appear to lose a larger number of rides than their competition, both in single-passenger and shared rides.

1. Introduction

This paper examines the effects of a surcharge implemented on taxis and for-hire-vehicle (FHV) programs in downtown Manhattan (NYC Taxi & Limousine Commission, 2023). The surcharge is designed to shift passengers into fewer vehicles and reduce overall traffic congestion, while at the same time be a revenue source for the city, estimated around \$1 million a day (Metropolitan Transportation Authority, 2021). This surcharge is widely considered a “Phase One” of a larger tax designed to limit vehicles in the same area (Ley, 2022), and so offers critical insight into the consequences for New York City (NYC) and the conjoined taxi and FHV industry. We find strong evidence suggesting that the modest 2019 surcharge, ranging between \$0.75 and \$2.75 per passenger, is nontrivial and reduces pickups in the affected region, and further evidence that riders are more likely to pool rides, i.e. travel together on the same vehicle, after the surcharge. While one might anticipate a larger decline in FHV usage, where the surcharge is larger, the overwhelming majority of this reduction in pickups is concentrated in taxis, specifically the so-called yellow cabs. NYC taxis system is split between yellow and green cabs, where yellow cabs are allowed pickups anywhere in the city, whereas green cabs can only pickup outside of the downtown Manhattan area (the area south of 96th St). Taxis are able to pick up passengers from the street spontaneously, called “street hailing”, and have extremely rigid pricing structures based on duration, destination, and travel time. Because of the complex pricing structure, it is all but straightforward estimating the final cost of a taxi trip ahead of time. FHV programs like Uber and Lyft, on the other hand, have seen a much smaller reduction in rides in the treated area and may have even gained some of the shared rides. FHV cannot street

hail, the trips are always prearranged, even if by a trivial duration, and prices are always known at time of pickup. In addition, we find significant evidence that even green cabs appear to have filled in for this vacancy in (or at least very near) the treatment area, despite the fact that such operations within the region are prohibited and heavily penalized by law.

Putting aside the inter-industry competition, regional safety appears to improve significantly within nearly every subset of collisions and injuries, and some categories of fatalities. Together with the decline in pickups, this suggests that reduced activity in the congestion zone is associated with improved safety. In turns, this supports the idea that future congestion taxes of larger scope and value may have a large impact. We find the suggestion of some potential spillover collisions in border regions on the northern portion of Manhattan, is consistent with the idea that a future congestion surcharge may have some spillover impacts with congestion.

Finally, under the new surcharge, cabs appear to increase the proportion of rides provided during rush hour after the surcharge. We find a significant increase in cab pickups during rush hours relative to off-peak hours within zones and time periods treated by the surcharge.

The remainder of this paper is organized as follows: in Section 2, we examine the relevant literature on congestion taxes and describe the background of the surcharge; in Section 3 we then examine the data available; in Section 4, we follow with a discussion of the methodology, a difference-in-differences approach; in Sections 5 and 6, we examine the primary impact of the surcharge on rides and then secondary impacts on safety and spatial and temporal spillover; finally, in Section 7

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we conclude with a summary of our results and direction for further research.

2. Background

2.1. Literature review

Congestion taxes have long been considered an essential tool for managing the externalities of traffic congestion. The earliest proposal for a congestion surcharge dates back to 1920 when Pigou proposes a tax on carts along a road (Button, 2020). Later, these problems continue to grow in prominence and Buchanan recognized the potential benefit to taxing automobiles in urban areas: “Our cities were simply not built to accommodate motor vehicle traffic of the magnitude now existing, and it is inconceivable to think of reconstructing them for this purpose”. Buchanan (1952). Similarly, Vickrey (1968) floats the idea of pricing for highway usage and the marginal contribution to accidents.

About a century later, these policies have been rapidly implemented in major urban areas and older cities (Lehe, 2019), including New York City (Board, 2022), Stockholm (Eliasson et al., 2009; Börjesson et al., 2022), Sao Paolo (Moita et al., 2017), London (Green et al., 2016), and Gothenberg (West and Börjesson, 2020). Congestion taxes are widely considered a first-best or near-best solution for improving roadway efficiency (Parry, 2002; Arnott and Small, 1994). The basic rationale is that roadways are a common resource prone to overuse unless drivers are charged for the space they occupy, which causes congestion for others.

Recent studies have highlighted the relationship between ridesharing programs and congestion. Barrios et al. (2020) found that the introduction of ridesharing programs (e.g., Uber, Lyft) is associated with increased vehicle-miles traveled, higher gas consumption, and additional annual hours of traffic delay. Notably, the authors observed a 3% increase in traffic fatalities, underscoring the link between road usage and accidents. A rough estimate suggests that the cost of these programs, purely in terms of traffic fatalities, ranges from \$5 to \$13 billion. These findings are not uncontested, however. Anderson and Davis (2021) find that ridesharing programs are associated with a 6.1% decline in alcohol-related fatalities and a 4% overall decline in fatalities — particularly concentrated in nights and weekends.

While licensing taxis to control the supply of rides is so prevalent that they are discussed in an exercise in Milton Friedman’s 1962 textbook (Friedman, 1962), taxation of for-hire vehicles (FHV) is a relatively new area of research. Lehe et al. (2021) provides a case study for the Illinois state government, summarizing the FHV taxation policies of several major cities and states, as well as the revenues gained from such policies. However, this report avoids examining the impact of these taxes on safety or congestion, focusing solely on revenue collection. This gap is addressed in our paper, which examines the broader effects of these taxes. Even without assessing these secondary impacts, Lehe et al. (2021) urged the Illinois state government to pass ride-hailing tax legislation. Uber, in a 2019 lawsuit against the Village of Skokie, IL, advocated for state-level taxation over local patchwork taxes, potentially making lobbying efforts easier while removing control from local governments, which are more directly affected by the consequences of such taxes. Notably, the surcharge discussed in this paper was imposed by the state of New York, not the city.

Lehe and Pandey (2022) present theoretical results suggesting that taxes or medallion (i.e., taxi license) costs may be necessary to achieve the socially optimal number of rides. Preliminary empirical results from Liang et al. (2022) indicate that FHV taxation reduces FHV demand by nearly 17%, with only a modest effect on traffic congestion, suggesting that FHVs make up a small portion of total traffic. However, they do not explore the consequences for road safety.

These anti-congestion policies are often unpopular upon announcement (Else, 1986), and the NYC congestion surcharge is no exception.

Table 1
Structure of surcharge.

Fee	Case
2.50	Non-shared trips in taxicabs
2.75	Non-shared trips in FHV
0.75	Shared rides in any type of vehicle
None	Access-A-Ride or any other MTA-dispatched trip

It was challenged in court by local taxi drivers (New York State Department of Taxation and Finance, 2019), which delayed the policy’s implementation. First, such policies impose additional taxes. Evidence and theory, e.g., Lehe and Devunuri (2021), suggest that individuals often go to great lengths to avoid tolls, even paying, to circumvent what was previously a “free” roadway. Second, concerns are raised about equity and the potentially regressive nature of these taxes (Peters and Kramer, 2012; Yuan et al., 2020; Karlström and Franklin, 2009). In Shefer and Rietveld (1997), authors argue that reducing roadway traffic yields diminishing returns in terms of safety, a theme further explored by Cappellari and Weber (2022) in the context of NYC. On the other hand, Vitaliano and Held (1991), argue that the externalities from additional road users are too small to be relevant.

Still, congestion taxes have been found to be associated with improved air quality (Green et al., 2020) and increased road safety (Singichetti et al., 2021). Drivers appear to gain in net welfare even without recycling revenues from these charges (Börjesson and Kristoffersson, 2014). These policies also offer the potential to reallocate resources to underutilized public transportation, as noted by Ley (2022). Overall, these studies suggest that despite some limitations, congestion taxes generally reduce traffic congestion (Börjesson and Kristoffersson, 2015, 2018) and contribute to improvements in safety and air quality.

Most similar to our work is the parallel (Li and Vignon, 2024), where we both use a difference in differences approach on the same intervention, but have 3 distinct differences. First, we use a much longer time frame to account for natural within-year variation: the smaller effect in Li and Vignon (2024) could be due to normal seasonal demand increases in the month following the treatment. These seasonal elements will not be controlled for without data from those dates previous years. Secondly, we also include the Brooklyn and Queens area as controls rather than only including Manhattan. We explore Manhattan-only estimates as a robustness check in Table 10. Third, they include market share of competing services as a control. This seems intuitive but is potentially hazardous – the amount of rides provided in a given hour does certainly depend on the number of competing services – but the competing services are also a consequence of the number of rides provided.

Our study is one of the first to examine the regional effects of the congestion surcharge on the transportation industry. This new surcharge, in many ways, responds to the relatively recent rise of urban ridesharing programs and serves as a “Phase 1” for the larger congestion taxes planned for the city (Fix NYC Advisory Panel, 2018; Ley, 2022). We are also one of the first to explore the competitive effects of the NYC surcharge on different sectors within the transportation industry (Li and Vignon, 2024).

2.2. NYC citywide regulations

New York City is no stranger to transportation regulation. Since the Hass Act of 1937, the number of cabs has been strictly limited by the issuance of medallions, taxi licenses, by the city (Cetin and Deakin, 2019). As a result, medallion prices have steadily increased along with the population. Previous analysis shows that a 1% decrease in the number of medallions leads to a 0.45% increase in medallion prices, which in turn drives up taxi fares (Cetin and Eryigit, 2013). Earlier efforts by government bodies to encourage carpooling had largely been unsuccessful (Cetin and Deakin, 2019; Deakin et al., 2010).

In 2011, Uber entered the NYC market, followed by Lyft in 2014. Given their similar role to traditional taxis, the entry of for-hire vehicle (FHV) programs significantly reduced both the taxi market share (Fischer-Baum and Bialik, 2015) and the value of medallions (Cetin and Deakin, 2019), a decline that has continued over time. Early research suggests that taxis improved the quality of their services in response to this competition (Wallsten, 2015).

In 2013, between Uber and Lyft's entry into the market, the Boro Taxi program, also known as the "Green Cab", was introduced. This program focuses on non-Manhattan areas and prohibits pickups in lower Manhattan, with fines of \$500 and suspension for the first offense for violations. Still, Green Cab drivers tend to operate near the downtown region, from which they are excluded (Barron, 2018). In 2018, immediately before the surcharge we are examining, no illegal pickups by Green Cabs were recorded in the restricted area, suggesting strong compliance. However, after the surcharge was implemented in 2019, there were several cases of illegal pickups in this area.

According to the NYC Taxi and Limousine Commission's (TLC) annual reports, the number of cabs and FHV's has remained relatively constant after the surcharge was imposed. This might be surprising, given several new rules were passed in 2018. Local Law 149 limited the number of FHV's and established additional requirements for licensees (New York City Taxi and Limousine Commission, 2023). At the same time, Local Law 148 paused the issuance of new FHV licenses. While one could assume that these measures could lead to dramatically reduce the number of cabs on the road, potentially accounting for improvements in safety, the New York City Taxi and Limousine Commission (2021) show that the number of cabs remained constant at 13,587 during our data window, while the number of FHV's peaked at about 136,000 when the regulation took effect in 2018.

Beyond cabs and FHV's, several other transportation options compete in NYC, most notably the subway, as well as bikes, walking, and even helicopters (Schillinger, 2017). The subway fare during this period was also set to \$2.75 per ride, aligning with the congestion surcharge, though discounted 10-ride packs and monthly passes are available for regular users. In 2019, subway fares increased, which may have driven some riders toward taxis and FHV's (Markham, 2022). Although this shift was citywide, if subway riders are concentrated in areas affected by the surcharge and are turning to cabs, our findings may be conservative.

Although our focus is not on these alternative systems, they often work in synergy with cabs and FHV's, which provide the "last mile" of transit (Hall et al., 2018). Funding for these competing public transit systems are also frequently cited as a justification for surcharges like the one under study (Ley, 2022).

2.3. Implementation of the congestion zone surcharge

The New York State congestion surcharge was initially scheduled for implementation on January 1, 2019, but was delayed by a court order until February 2, 2019. This surcharge specifically targets for-hire vehicles (FHV's) and taxis, aiming to reduce congestion in downtown Manhattan. The policy was a major component of the governor's 2019 budget (NY State - Division of the Budget, 2018) and received significant attention in local news outlets (Spectrum News, 2018), making it widely known to the public. The surcharge is clearly listed on all receipts. The law states that it "cannot be taken out of the driver's pay and must be passed on to the passengers". The funds are collected by the technology service provider or transportation network company (usually the app which collects the trip record — like Uber, Lyft, or the other associated contractors who manage taxi billing and ride tracking) and rendered to NYS Department of Taxation and Finance monthly. Failure to pay results in fines of up to 200% of the amount due in addition to other penalties and interest.

The surcharge applies to the downtown Manhattan area, as shown in Fig. 1 (NYC Taxi and Limousine Commission, 2019). As illustrated

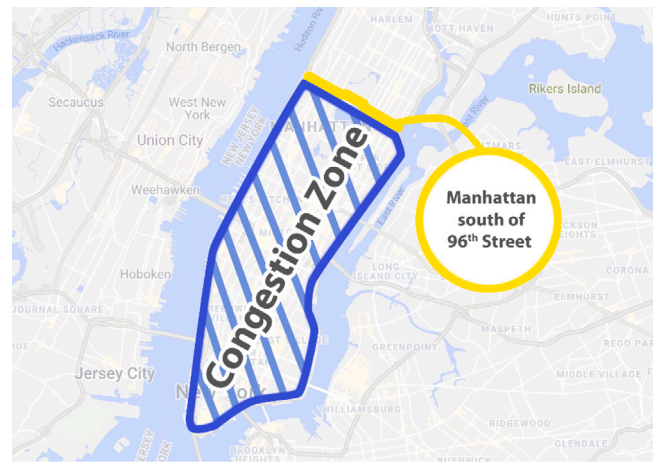


Fig. 1. Area targeted by the surcharge, official document from NYC.gov.

in the figure, the surcharge, or congestion, area is defined as the part of the Manhattan island that is south of the 96th Street. The 96th street is the northern-most border of the surcharge area, with the remaining borders being defined by the island coastal line south of such street. The fee structure is designed to encourage shared or "pooled" rides by charging differential rates, as outlined in Table 1 (NYC Taxi & Limousine Commission, 2023). Pooled rides are defined as those rides where multiple passengers share a vehicle, or where a rider indicates willingness to share their ride with another passenger in the app. In this paper, we will interchangeably use the terms *shared* and *pool* rides to refer to such rides. Similarly, we will use the terms *single ride* or *single trip* to refer to rides that are not pooled, that is rides with only one passenger.

The largest surcharge, \$2.75 (equivalent to a subway fare), applies to FHV trips within the designated area. Taxi rides within the same area incur a slightly lower surcharge of \$2.50. For street-hail taxi rides with a single passenger, the full surcharge applies. However, for pooled rides the fee is reduced to \$0.75 per passenger. We presume this price differential is to motivate shared rides where multiple passengers are in the same vehicle instead of separate ones, eg the same as carpooling. Importantly, a ride is classified as pooled based on the rider's willingness to share, not the actual number of passengers. Therefore, a pooled trip might still have only one passenger if no one else joins the ride. All rides scheduled through apps like Uber are considered pre-scheduled, even if the time between booking and pickup is very short.

Before delving into the main analysis, we examined data to confirm that the surcharge area aligns with the region experiencing the highest concentration of collisions. Fig. 2 shows that the majority of collisions occur in lower Manhattan, which is precisely where the surcharge is applied. These surcharge zones tend to have smaller land areas and higher population densities. Along with collision data, Fig. 2 also shows the taxi zones across all boroughs, indicated by the red lines which define the borders between zones.

While this correlation suggests that the surcharge could potentially reduce collisions in these high-traffic areas, it alone does not guarantee that the surcharge will be effective or sufficient in achieving this goal.

3. Data

We collected data on two main sources: rides (New York City Taxi and Limousine Commission, 2022) and motor vehicle collisions (New York City Police Department, 2022). Unfortunately, we do not have access to data on traffic congestion or vehicle idling times, which might have provided a fuller picture of traffic patterns. Additionally, publicly available traffic volume data, such as the toll data used in studies

Table 2
Summary of ride data.

	Non-Surcharge zones		Surcharge zones		All zones	
	Mean/Zone	SD	Mean/Zone	SD	Mean/Zone	SD
Total trips, Single cab (Yellow)	119	573	3460	2413	746	1750
Total trips, Shared cab (Yellow)	47	238	1376	928	296	691
Total trips, Single cab (Green)	95	230	5	31	78	211
Total trips, Shared cab (Green)	17	43	1	6	14	39
Total trips, Single FHV	1464	2462	4113	2293	1962	2642
Total trips, Shared FHV	122	209	229	254	142	222
Total passengers, Shared cab (Yellow)	146	746	4342	2942	934	2183
Total passengers/Shared cab (Yellow)	3.10	0.90	3.15	0.10	3.12	0.78
Total passengers, Shared cab (Green)	56	144	3	21	46	132
Total passengers/Shared cab (Green)	3.26	0.93	3.54	0.92	3.27	0.93
Total collisions	2.280	2.252	1.683	1.636	2.168	2.162
Total cars in collision	4.407	4.433	3.222	3.190	4.185	4.252
Total injuries	0.651	1.240	0.302	0.698	0.585	1.166
Total deaths	0.003	0.052	0.002	0.052	0.002	0.052
Total fares, Single cab (Yellow)	\$3,600.62	\$25,559.20	\$51,846.85	\$36,342.35	\$12,661.44	\$33,670.15
Total fares, Shared cab (Yellow)	\$1,506.50	\$11,259.37	\$21,126.51	\$14,876.52	\$5,191.21	\$14,256.38
Total fares, Single cab (Green)	\$1,423.89	\$3,135.96	\$77.61	\$443.12	\$1,171.05	\$2,881.09
Total fares, Shared cab (Green)	\$259.38	\$610.96	\$13.58	\$81.51	\$213.22	\$560.03
Average fare/Single cab	\$24.93	\$47.31	\$15.88	\$8.93	\$23.09	\$42.57
Average fare/Shared cab passenger	\$8.56	\$8.02	\$5.14	\$1.18	\$7.79	\$7.22

N: 290,175; i = 265 (49 treated, 207 untreated); t = 1095.

Zones with no fares are omitted from fare estimation. Totals are daily sums.

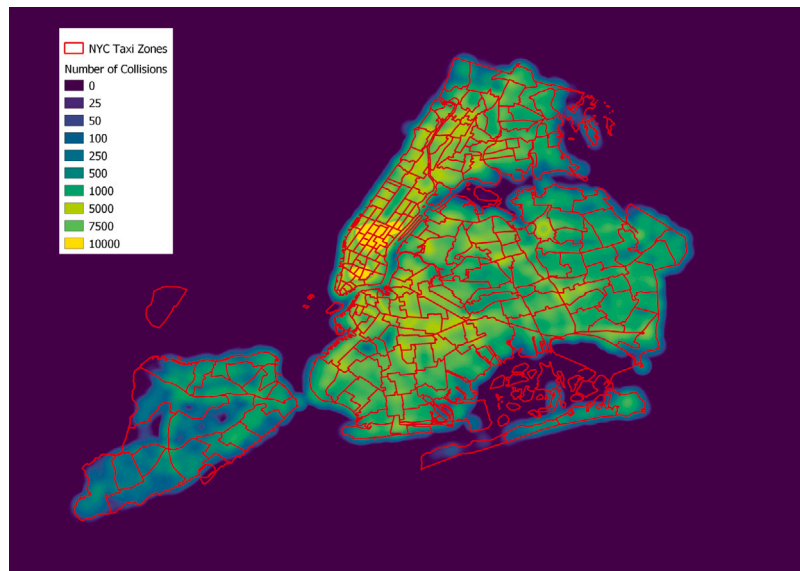


Fig. 2. Hot spots of collisions and taxi regions.

like Cappellari and Weber (2022), is not fully applicable to downtown Manhattan traffic, since much of the movement is internal to the area and not subject to tolls.

In this paper, we focus the analysis on assessing the impact of the surcharge hike implemented in New York City in early 2019. Our dataset for the analysis extends to ride data between January 2017 and December 2019. This timeframe includes two years before the surcharge and one year after, allowing us to observe trends and changes pre- and post-surcharge.

In the remainder of this section, we provide a description of the data sets, and we provide important insights into city-wide transportation patterns and safety.

3.1. Rides data set

This dataset captures information on taxi and for-hire vehicle (New York City Taxi and Limousine Commission, 2022) rides in New York City. Data has been collected by the Taxi and Limousine Commission

(TLC) for many years as part of the NYC Open Data initiative (Open-Data, 2020). The rides data spans from 2009 to the present day and covers all types of vehicle offering transportation services, including yellow cabs, green cabs, and ride-hailing services such as Uber and Lyft. One key challenge in using this dataset is the difference in structure between the taxi and FHV data. The taxi data set is quite detailed. It includes attributes describing the pickup and dropoff locations and times, the number of passengers, the distance traveled, the payment type (cash or card), the amount paid by the customers. In particular, the amount paid is provided with its breakdown, which includes: the total amount charged, minus any tip; the fare amount, that is, the time-and-distance fare calculated by the meter; the tip, if payment is made by card (cash tips are not automatically tracked); toll amounts; any extras, such as luggage; and a number of taxes: the MTA tax, the improvement surcharge, and the congestion surcharge. The FHV dataset contains rather limited information. This data set reports only on pickup and drop-off locations and times, and whether the ride was a shared, pool, ride. Additional data like travel distance or itineraries would enable

much finer analysis. Unfortunately, no further data is provided in either data sets.

In order to conduct our analysis, we had to address these discrepancies. We created a unified dataset that merges on the common attributes across both taxis and FHV's datasets, namely the pickup and dropoff locations and times. This process we do not discard attributes or values: the unified dataset contains all data, whereby null values are introduced in the merged data set for those attributes that are not present in the original records. This process creates a somewhat sparse dataset of individual rides. We consolidate such data set by creating an aggregated summary view of all rides for each day throughout the time interval of interest. Specifically, the data are aggregated by day and by taxi zones, where each record provides summarize key information such as: the hour of the day, the pickup zone, and the number of pickups per vehicle type (yellow cabs, green cabs, or FHV's) and trip type (single ride or pooled ride).

Some FHV researchers may anticipate exact latitude and longitude coordinates for pickup locations, which existed in the past and allowed for precise mapping. However, due to privacy concerns about tracking rider behaviors, the city revised its data collection policy, and now this level of detail is no longer provided. As a result, the current FHV data reports pickups from broader taxi zones.

3.2. Motor vehicle collision data set

The second dataset tracks motor vehicle collisions in New York City ([New York City Police Department, 2022](#)). It is maintained by the New York Police Department (NYPD) and made available through the NYC Open Data initiative ([OpenData, 2020](#)). Under New York City law, collisions must be reported if they result in injury, death, or at least \$1000 in property damage. This ensures that the dataset includes a reasonably complete record of all significant traffic incidents, not just those related to taxis or FHV. Each entry in the collision dataset contains detailed information about the crash, including the date and time it occurred, its location, the number of people injured or killed (categorized into motorists, cyclists, and pedestrians), contributing factors (e.g. following too closely, driver distraction, failure to yield right-of-way, etc.), and the types of vehicles involved (e.g. Sedan, station wagon, SUV, truck, etc.). However, to protect privacy, sensitive information that could identify the individuals or vehicles involved is excluded. This collision data provides a valuable resource for understanding the relationship between transportation patterns and traffic safety. It helps us examine whether the surcharge, which likely altered traffic flow and ride behaviors, had any measurable effect on the frequency or severity of traffic collisions across the city.

As detailed in [Table 2](#), there are noticeable differences between areas affected by the surcharge and those that are not. Surcharge regions are typically high-traffic areas, experience higher density, more rides, and greater transportation demand. These areas account for a larger share of the total rides, and correspondingly, generate more revenue from fares. In contrast, non-surcharge areas see fewer rides overall. However, when rides do occur, they tend to cover longer distances and result in larger fares. Despite the lower frequency, the overall fare amounts in non-surcharge areas can still be significant due to the distance traveled. From a safety perspective, [Table 2](#) shows less variation between the surcharge and non-surcharge regions. The number of collisions, injuries, and other safety-related incidents tends to fall within a standard deviation of each other across these areas. This suggests that while the surcharge may have shifted some traffic patterns, it is not immediately obvious if it is associated with improved traffic safety outcomes.

By focusing on these two datasets, rides and motor vehicle collisions, we are able to investigate the effects of NYC's surcharge hike on both transportation behaviors and public safety. The ride data helps us understand how the number and type of trips evolved over time, while the collision data provides insights into any changes in traffic incidents during the same period. Together, these datasets offer a comprehensive view of the broader impacts of transportation policies on the city's traffic and safety landscape.

Table 3

Initial changes in pickups after treatment in treatment period.

	All pickups	Yellow	Green	FHV
DiD	-1103.745*** (-5.15)	-1181.299*** (-11.56)	51.995*** (5.73)	264.621 (1.10)
Pct change (%)	-11.6	-24.2	752.5	5.7
R^2_a	0.866	0.944	0.860	0.694
N	290 175	290 175	290 175	290 175

Clustered by taxi zone, *t*-statistics shown in parentheses.

Significance levels are: *** = 0.1%, ** = 0.1%, * = 5%.

Percentage change is $\beta/\bar{y}$ for the average treated regions.

4. Methodology

In this section, we outline the steps taken to investigate the causal relationship between the congestion surcharge and its impact on taxi pickups and collision rates. Our goal is to determine if there is a plausible connection between the introduction of the surcharge and the observed decline in collisions, using robust econometric techniques.

[Fig. 3](#) offers a preliminary exploration of taxi pickup trends in areas affected by the surcharge compared to non-surcharge areas. This aggregate analysis serves as the foundation for our more detailed empirical investigation. A key observation is the strong seasonality in taxi pickups in the treated areas, particularly an increase in pickups during the last quarter of the year. This seasonal pattern is essential to consider in our analysis, as it could introduce bias if not properly accounted for. Therefore, we include daily fixed effects across all 265 taxi zones in our regression models. These fixed effects allow us to control for unobserved time-invariant factors that might affect taxi pickups, such as local characteristics of the zones.

Our primary interest lies in the change in taxi pickups in surcharge-affected areas relative to the nearly unchanged trends in untreated areas. This differential effect forms the crux of our analysis, as it helps isolate the impact of the surcharge from other confounding factors that may affect taxi pickups over time.

To quantify the effect of the surcharge on pickups, we employ a difference-in-differences (DiD) approach. The DiD method is widely used to estimate causal effects when a policy or treatment is introduced in some areas (treated) but not in others (control). By comparing the changes in outcomes in treated areas before and after the surcharge to those in untreated areas, we can estimate the causal impact of the surcharge. We further refine this analysis by decomposing rides into single and pooled categories. This disaggregation allows us to explore potential heterogeneity in the response to the surcharge across different types of rides and services, providing a more nuanced understanding of the policy's effects.

We have two possible mechanisms that might relate the surcharge to collisions. The first is that vehicles move into an area in order to perform pickups, and reduced pickups reduce the number of vehicles traveling in this area. Reduced vehicle counts result in fewer collisions ([Dickerson et al., 2000](#); [Cappellari and Weber, 2022](#)). The second is that perhaps the pickup activity narrows lanes in an already congested region — resulting in lane and speed changes that can be potentially hazardous relative to the ideal homogeneous traffic flow ([Tang et al., 2009](#); [Sugiyama and Nagatani, 2012](#)). There is already a preexisting body of literature on double parking ([Kim and Wang, 2022](#); [Han et al., 2005](#)), freight pickups/drop-offs ([Conway et al., 2016](#)), and their respective potential hazards. However, all of these arguments hinge on pickups changing within the region. If the change in pickups of all types are too small to measure, then there is no reason to expect vehicle counts to change either. Similarly, if the pickup activity itself is hazardous, but remains unchanged, then the point is moot. In either case, we have a necessary obligation to demonstrate that pickups change within the region.

One might hypothesize that instead of reducing pickups, taxis could remain in the area and drive without passengers. In such cases, we

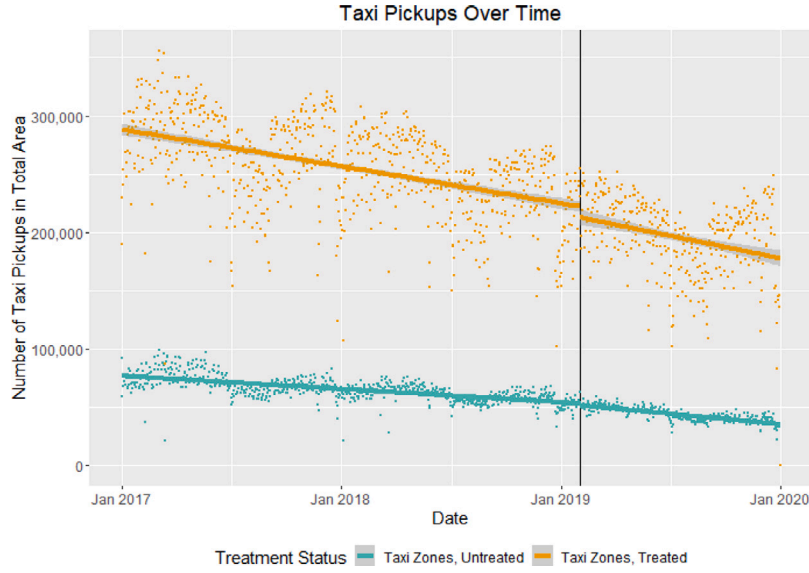


Fig. 3. Parallel trends for taxi industry.

would not expect to see a substantial reduction in collisions, as vehicle presence on the roads would remain unchanged. To address this possibility, we specifically focus on the relationship between the number of pickups and subsequent collision rates.

Eq. (1) models the effect of the surcharge policy to taxi pickups.

$$Rides_{i,t} = \alpha_i + \delta_t + \beta_1 SurchargeApplies_{it} + \epsilon_{i,t} \quad (1)$$

In this equation, $Rides_{i,t}$ represents the number of pickups in zone i on day t . The fixed effects for each zone are captured by α_i , controlling for time-invariant factors specific to each zone. Daily fixed effects, δ_t , account for time-specific factors such as weather or holidays that may affect pickups across all zones. The key coefficient of interest is β_1 , which measures the impact of the surcharge on the number of rides. Specifically, it captures the relative change in rides for zones where the surcharge applies during the surcharge period.

To better understand the nature of this decline, we repeat the estimation for different subsamples. These subsamples are defined by service type (yellow cabs, green cabs, and FHV services) and ride type (single and pooled rides). This decomposition allows us to explore whether certain services or ride types are more affected by the surcharge than others. For example, pooled rides may be more price-sensitive than single rides, leading to a larger reduction in pickups for pooled services.

After defining how to measure ridership behavior around pre and post the surcharge implementation, we move to the next step in our analysis: determining whether this decline correlates with a reduction in collisions. Using a similar DiD framework, Eq. (2) models the impact of the surcharge on collision rates.

$$Collisions_{i,t} = \alpha'_i + \delta'_t + \beta'_1 SurchargeApplies_{i,t} + \epsilon'_{i,t} \quad (2)$$

In this model, $Collisions_{i,t}$ represents the number of collisions in zone i on day t . The fixed effects α'_i and δ'_t serve the same purpose as in the previous model, controlling for zone-specific and time-specific factors, respectively. The coefficient β'_1 measures the effect of the surcharge on collision rates, allowing us to assess whether the policy has contributed to a safer environment in the treated areas.

To provide a more comprehensive analysis, we decompose the total number of collisions by both severity (e.g., fatalities, injuries) and participant type (pedestrians, cyclists, motorists). This breakdown allows us to determine whether the surcharge has differential effects on different types of road users or on more severe collisions. For instance, it is possible that the surcharge disproportionately reduces collisions

involving pedestrians or cyclists, who are more vulnerable in urban traffic environments.

A critical assumption of the DiD approach is the parallel trends assumption, which posits that in the absence of the surcharge, the treated and control groups would have followed similar trends over time. Although this assumption cannot be directly tested, we conduct an event study analysis to assess its plausibility. Specifically, we include one-month lags and leads around the introduction of the surcharge, covering 17 months before and 11 months after the policy's implementation. The exclusion of the period immediately before the surcharge as a base allows us to isolate the policy's effects more clearly.

In addition to examining the overall effects of the surcharge, we consider whether changes in ridership differ by time of day. Morning and evening rush hours, in particular, may see different patterns of ridership and congestion due to higher demand for taxis during these periods. Taxis may also behave differently during rush hours because longer trips, which are more common during peak times, reduce the proportional impact of the surcharge. To account for these potential variations, we employ a triple-difference-in-differences (DiDiD) estimator, which extends the DiD approach by introducing an additional dimension of analysis: time of day. The DiDiD estimator is more robust to violations in parallel trends. DiDiD can obtain unbiased estimates without parallel trends — so long as the pair of sub-DiD estimates being compared have the same bias (Olden and Møen, 2022). This estimator also provides a clearer picture of how the surcharge affects behavior during different times of the day. We define rush hours as 7:00–8:59 am in the morning and 5:00–6:59 pm in the evening, while off-peak hours are defined as 9:00 am–4:59 pm and 7:00 pm–6:59 am. The model for this analysis is specified in Eq. (3).

$$Rides_{it} = \alpha_i + \delta_t + \beta_1 surchargeGroup_i * surchargePeriod_t + \gamma_j rushHour_j + \beta_2 surchargeGroup_i * rushHour_j + \beta_3 surchargePeriod_t * rushHour_j + \beta_4 surchargeGroup_i * surchargePeriod_t * rushHour_j + \epsilon_{it} \quad (3)$$

The key coefficient of interest here is β_4 , which captures the interaction between the surcharge and rush hours. If β_4 is significant, it would indicate that the impact of the surcharge on taxi pickups differs between rush and non-rush hours, potentially leading to increased congestion during critical periods.

In summary, by using a combination of DiD and DiDiD estimators, our methodology accounts for both temporal and spatial variations in

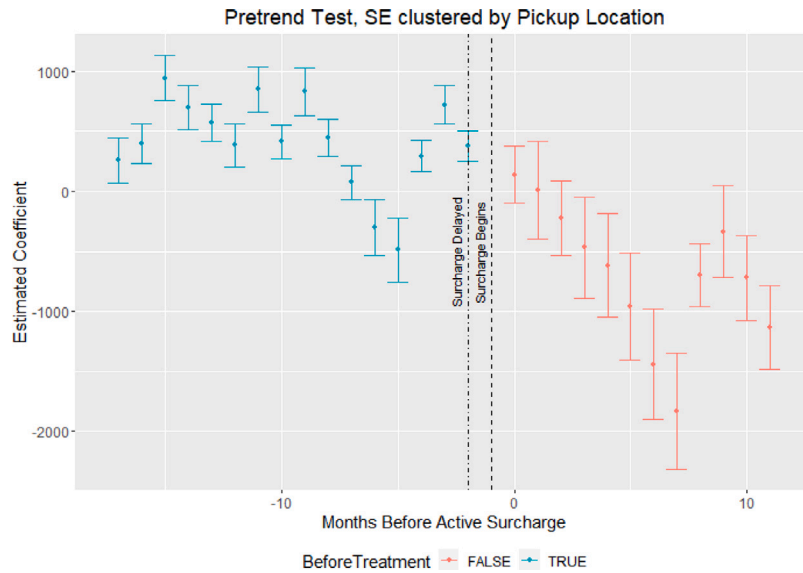


Fig. 4. Pretrend test for difference in differences.

taxi pickups and collision rates. This approach allows us to rigorously estimate the causal impact of the congestion surcharge, while also providing insights into how the policy affects different types of rides, services, and times of day.

5. Results

The first result of our study on the effects of the surcharge introduced in lower Manhattan on different types of taxi and ride-hailing services is that the surcharge had a notable impact on the volume of pickups in the treated areas. Fig. 4 presents the results of an event study analysis of total ridership patterns. We observe a seasonal trend in the data, with pickups in the treated areas increasing during the last quarter of the year, likely due to the holiday season. However, following the implementation of the surcharge, ridership in the treated areas declined substantially compared to control areas. While there was some partial recovery in the final quarter of the year, the data provide clear evidence of an overall decline in ridership in the treated zones after the surcharge took effect.

5.1. Inter-service competition

In this section, we analyze the impact of the surcharge by each sub-type of ride. Table 3 presents the results of applying Eq. (1) to our collected data partitioned by ride type. In Table 3, column 1 highlights the significant reduction in pickups following the introduction of the surcharge, with about 1104 fewer pickups per day in each affected taxi region. This represents an 11% decline compared to the pre-surge average number of rides, which was approximately 9606 pickups per day in these regions. The introduction of the surcharge appears to have effectively discouraged a significant portion of rides in these zones.

The decrease in pickups was primarily driven by a reduction in yellow cab rides. As seen in column 2 of the table, yellow cabs accounted for the majority of this decline. The reduction in yellow cab pickups is particularly concerning for the traditional taxi industry, as yellow cabs have historically been a primary mode of transportation in Manhattan. However, this decline was partially offset by a rise in pickups by green cabs, as noted in column 3. Despite the fact that green cabs are legally restricted from picking up passengers in lower Manhattan – subject to a \$500 fine and suspension on the first offense – we observed a small but statistically significant increase in green cab pickups in this zone. During the surcharge period, we observe the first

Table 4

Initial changes in pickups after treatment in treatment period by single/shared rides.

	Single rides		Shared rides	
	Cab	FHV	Cab	FHV
DiD	−805.839*** (−10.87)	−261.519 (−1.19)	−318.659*** (−11.08)	252.216*** (12.46)
Pct change (%)	−23.3	−6.8	−23.2	30.6
R^2_a	0.936	0.646	0.945	0.856
N	290 175	242 210	290 175	242 210

Clustered by taxi zone, t-statistics shown in parentheses.

shared ride data is not available for FHV's prior to July 2017.

Significance levels are:*** = 0.1%, ** = 0.1%, * = 5%.

Percentage change is $\beta/\bar{y}$ for the average treated regions.

green cabs being prosecuted for this charter violation, suggesting this replacement is actually occurring and is not just an artifact of the data.

Additionally, we observed a slight decline in pickups by FHV's, i.e. Uber and Lyft. This decline, however, was small and statistically insignificant, suggesting that FHV's were not as affected by the surcharge as traditional taxis. This finding implies that FHV services may have some insulation from the impact of the surcharge, possibly due to their pricing flexibility, in contrast to taxis which cannot alter their rates.

While the overall number of pickups declined, as Table 3 shows, the effect was most pronounced for yellow cabs, whereas FHV's and green cabs exhibited different dynamics. The introduction of the surcharge led to a noticeable shift in competition between different types of ride services, with yellow cabs bearing the majority of the burden. Yellow cab pickups fell by approximately 11%, which is 1130 rides per day in each affected zone, amounting to a loss of about 55,000 yellow cab rides per day across the entire treated area. This sharp decline highlights the vulnerability of the yellow cab industry to policy changes like the congestion surcharge, which disproportionately affects services that rely on street hails and traditional taxi meters. Interestingly, despite the negative impact on yellow cabs, Uber – a leading FHV provider – publicly expressed cautious support for the surcharge. This is an unusual stance for a company whose primary revenue source is directly taxed by the policy. In fact, Uber has historically resisted any regulation or taxation that could affect its pricing structure. However, in this case, Uber's support may reflect a strategic calculation that the surcharge would harm traditional taxis more than FHV's, thereby increasing its market share (Gold et al., 2022).

One key point to consider is that our pricing data only covers yellow and green cabs, not FHV's. Yellow cabs have a fixed pricing

Table 5

Simultaneous reduction of traffic and collisions in taxed regions in surcharge periods compared to all boroughs.

	All pickups	Collisions	Cars in collisions
DiD	-1103.745*** (-5.15)	-0.087** (-2.67)	-0.164* (-2.59)
Pct change (%)	-11.6	-5.2	-5.1
R^2_a	0.866	0.498	0.483
N	290 175	290 175	290 175

Clustered by taxi zone, t-statistics shown in parentheses.

Significance levels are:*** = 0.1%, ** = 0.1%, * = 5%.

Percentage change is $\beta/\bar{y}$ for the average treated regions.

structure determined by meters, whereas FHV's have more dynamic pricing models that can fluctuate based on demand, time of day, and other factors. The \$0.25 difference in the surcharge between yellow cabs and FHV's may not be enough to offset the competitive advantage that FHV's gain from their flexible pricing. Moreover, there are several other potential factors that could explain why FHV's did not experience the same level of ridership decline as yellow cabs. For instance, FHV's may have different customer demographics or a position as the default service that insulates them from the consequences of price increases.

5.2. Shift to pooled rides

Another important aspect of the surcharge policy is its differential impact on single versus pooled rides. The surcharge is structured to discourage single rides and incentivize pooled rides by making the latter cheaper, both on a per-passenger and total cost basis. The logic behind this approach is simple: by encouraging more passengers to share rides, fewer vehicles will be on the road, thereby reducing congestion in the city.

Table 4 examines whether this shift from single to pooled rides actually occurred. Using the same difference-in-differences (DiD) methodology described earlier (see Eq. (2)), we analyze ridership patterns for both single and pooled trips across taxi and FHV services. It is important to note that FHV passenger data is only available from July 1, 2017, onward, so the analysis is limited to this period for FHV's.

Our findings reveal a significant decline in single-rider cab trips in the treated areas after the surcharge, with approximately 806 fewer single-rider trips per day per taxi region, a 23% decrease. There was also a reduction in pooled taxi trips, with about 319 fewer pooled rides taxi region per day, a 23.3% drop. While the numerical decline was larger for single-rider trips, the proportional reduction was similar for both types of rides.

For FHV's, the surcharge was higher, so we expected to see a greater shift toward pooled rides. Indeed, the data confirm this expectation. Single-rider FHV trips decreased by about 260 rides per day per region (a 7% drop), while pooled FHV trips increased significantly by about 250 rides per day (a 31% rise). This shift highlights the effectiveness of the surcharge in encouraging ride-sharing, particularly within FHV services.

One key difference between taxis and FHV's is the ease with which passengers can arrange pooled rides. FHV platforms allow users to book pooled rides with strangers — through an easily available menu option, and often at an already discounted price. These menu option also automatically counts the ride as “pooled” for pricing purposes even if no other rider is added to the trip. This menu option reduces the friction associated with coordinating pooled trips as compared with street hailing taxis and pooling with friends at the point of pickup. We suspect this feature likely contributed to the greater shift toward pooled rides in FHV services compared to traditional taxis.

In summary, Table 4 illustrates the substitution effect from single to pooled rides following the surcharge, particularly within the FHV sector. While the policy appears to have successfully promoted ride-sharing, this shift came primarily at the expense of the traditional taxi industry, which experienced a more substantial decline in ridership.

6. Pickups and safety

After establishing that pickups in the treated area have significantly declined following the congestion surcharge, we turn our attention to the question of safety. Specifically, we examine whether the reduction in transportation activity is associated with a corresponding decrease in traffic collisions. Our findings, presented in Table 5, provide evidence that the introduction of the congestion surcharge is associated with a meaningful reduction in both transportation activity and traffic-related collisions within the treated areas. This result suggests that the surcharge, while designed primarily to manage traffic and reduce congestion, may have also contributed to enhanced safety conditions in the region.

In Table 5, column 1 reiterates the significant decline in taxi pickups, which amounts to about 1104 fewer pickups per treated region per day. This reduction in transportation activity suggests the surcharge was sufficient to induce changes in FHV and taxi behavior. Column 2 reports a significant decline in traffic collisions, with approximately 0.087 (5%) fewer collisions per zone per day, which amounts to around 4.3 fewer collisions citywide. Column 3 further corroborates the previous column by showing a significant reduction in the number of cars involved in traffic accidents, with a decline of 0.164 (5%) cars per zone per day, or about 8 fewer cars across the city. On average, this suggests that fewer cars on the road directly contribute to a reduction in collisions.

These reductions in traffic incidents are consistent with the goals of the congestion surcharge policy, which seeks to reduce vehicular activity in congested areas. Importantly, the data demonstrates that the policy not only achieves a significant reduction in vehicle pickup activity in the area, significantly increases shared rides while reducing individual rides, but also is associated with fewer road accidents, potentially saving lives and preventing injuries. In the remainder of this section we provide further discussion on how the new policy impacted: the change in the number of casualties from collisions, the traffic patterns around the border of the policy enforcement area, and the congestion during rush hours.

6.1. Casualties from collisions

Table 6 breaks down the data on casualties resulting from traffic collisions, categorizing them into injuries and fatalities. Injuries represent the majority of traffic casualties, while fatalities are rare. By estimating the changes in injuries and fatalities in the treated areas during the surcharge period, we can assess whether the decline in transportation activity has had a tangible impact on public safety.

Column 1 of Table 6 shows a significant decline in injuries, with about 0.026 (9%) fewer injuries per zone per day. Over the course of a year, this amounts to an estimated reduction of approximately 465 injuries. Column 2 reveals an insignificant but notable decline in fatalities, with about 0.0004 (44%) fewer fatalities per zone per day, which equates to around 7 fewer fatalities per year. While the decline in fatalities is not statistically significant, it suggests a positive trend in public safety due to the reduction in traffic activity.

The data also allows us to categorize casualties by the type of vehicle involved in the collision. In columns 3 and 4, we observe insignificant declines in pedestrian injuries and fatalities, indicating that the congestion surcharge has not disproportionately affected pedestrian safety. Similarly, in column 5, there is an insignificant decline in cyclist injuries, but column 6 shows a small yet significant reduction in cyclist fatalities. For motorists, the decline in injuries, column 7, is insignificant, while no significant change is observed in motorist fatalities, column 8. Collectively, these findings indicate that the reduction in transportation activity has led to a cohesive improvement in safety, with no significant increases in injuries or fatalities in any disaggregated category.

Table 6
Casualties by type.

	Injured	Fatalities	PedInjured	PedFatal	CycleInjured	CycleFatal	MotorInjured	MotorFatal
DiD	-0.026** (-2.67)	-0.000 (-0.78)	-0.004 (-1.07)	-0.000 (-0.20)	-0.002 (-0.86)	-0.000 (-1.95)	-0.020** (-2.60)	0.000 (0.11)
R^2_a	0.174	0.001	0.068	0.001	0.050	0.000	0.142	0.001
N	290 175	290 175	290 175	290 175	290 175	290 175	290 175	290 175

Clustered by taxi zone, *t*-statistics shown in parentheses.

Significance levels are: *** = 0.1%, ** = 0.1%, * = 5%.

Table 7
Robustness to spillover and Ill-Defined treatment regions (Pickups)

	Including border interactions	Omitting Half-Treated	Omitting both
Border X SurchargePeriod	-161.812 (-1.26)	-161.812 (-1.26)	
HalfTreated x SurchargePeriod	-672.221* (-2.19)		
DiD	-904.872*** (-9.57)	-904.872*** (-9.57)	-904.872*** (-9.57)
Const	3402.935*** (594.97)	3365.513*** (592.48)	3374.315*** (616.83)
r^2_a	0.935	0.935	0.935
N	284 700	282 510	270 465

Clustered by taxi zone, *t*-statistics shown in parentheses.

Significance levels are:*** = 0.1%, ** = 0.1%, * = 5%.

Table 8
Robustness to spillover and Ill-Defined treatment regions (Collisions).

	Including border interactions	Omitting Half-Treated	Omitting both
Border X SurchargePeriod	0.026 (0.46)	0.026 (0.46)	
HalfTreated x SurchargePeriod	-0.096 (-1.25)		
DiD	-0.077* (-2.29)	-0.077* (-2.29)	-0.077* (-2.29)
Const	2.177*** (1002.54)	2.182*** (1014.65)	2.201*** (1124.49)
r^2_a	0.490	0.490	0.491
N	284 700	282 510	270 465

Clustered by taxi zone, *t*-statistics shown in parentheses.

Significance levels are:*** = 0.1%, ** = 0.1%, * = 5%.

Overall, the data presents a compelling case for the effectiveness of the congestion surcharge in improving road safety. While the primary goal of the policy was to reduce traffic congestion, the reduction in injuries and fatalities suggests that fewer vehicles on the road lead to safer streets for all road users.

6.2. Spatial spillover and stable treatment assumptions

In this section, we consider whether the reduction in pickup activity and collisions observed within the treated area has simply been displaced to areas bordering the surcharge zone. If traffic has shifted to adjacent areas rather than being eliminated, then it could undermine the effectiveness of the policy and disrupt the stable treatment assumptions underlying our analysis.

To explore this possibility, [Tables 7](#) and [8](#) present analyses of traffic displacement. We define a “border zone” as any taxi zone that is adjacent to the treated area but not itself subject to the surcharge. Additionally, some zones are only partially treated by the surcharge, referred to as “half-treated zones”. These zones could potentially experience increased traffic if vehicles are attempting to avoid the surcharge by operating just outside its boundaries.

In [Table 7](#), column 1 tests for an association between the surcharge period and an increase in pickups in the border zones. If traffic is merely being displaced, we would expect to see a significant increase in pickups in these adjacent zones. However, the results indicate an

insignificant decline of 162 pickups per day in the border zones, suggesting that traffic has not shifted to these areas. In fact, the half-treated zones also experience a significant decline in pickups, indicating that the surcharge is effectively reducing traffic activity even in these marginally affected areas. Columns 2 and 3 further support these findings by showing that, after removing the half-treated and border zones from the analysis, the decline in pickups within the treated areas remains significant. Even after accounting for potential spillover effects, we continue to observe a reduction of over 900 pickups per day per zone.

Turning to traffic collisions, [Table 8](#) tests for a similar displacement effect in the number of collisions. Again, we find no significant increase in collisions in the border zones, suggesting that the reduction in traffic activity within the treated area has not simply shifted the risk of collisions to adjacent areas. The half-treated zones also experience an insignificant decline in collisions, consistent with the reduction in pickups. After excluding these zones from the analysis, the decline in collisions within the treated area remains significant, with a reduction of 0.077 collisions per zone per day, similar to the original estimate of 0.087.

6.3. Rush hour

A potential concern with the congestion surcharge is that it might have different effects on transportation activity during rush hours

Table 9
Exploration of rush hour times.

	Cabs	FHV
Rush hour	667.370*** (10.45)	-35.951 (-0.26)
r_a^2	0.845	0.605
N	580 350	580 350

Significance levels are:*** = 0.1%, ** = 0.1%, * = 5%.

Table 10
Collisions by borough subsets.

Boroughs	Manhattan	Manhattan Brooklyn	Manhattan Bronx	Manhattan Brooklyn Bronx	Manhattan Brooklyn Bronx Queens
DiD	-0.126 (-1.90)	-0.113** (-3.10)	-0.116** (-3.21)	-0.112*** (-3.51)	-0.138*** (-4.65)
Const	1.759*** (117.76)	2.241*** (526.30)	1.896*** (382.83)	2.210*** (786.06)	2.280*** (1221.27)
r_a^2	0.401	0.503	0.401	0.480	0.485
N	72 211	138 945	118 425	185 159	259 212

Clustered by taxi zone, t-statistics shown in parentheses.

Significance levels are:*** = 0.1%, ** = 0.1%, * = 5%.

compared to off-peak periods. Rush hours, when traffic is typically at its heaviest, might present a unique challenge for both taxis and for-hire vehicles (FHVs). Because cabs are paid based on the duration of the trip, they may be incentivized to operate more heavily during rush hour when traffic is slow, allowing them to collect more revenue even after accounting for the surcharge. At the same time, street-hailing is most likely to occur during these periods, as individuals need quick, on-demand transportation in congested business districts.

To test this hypothesis, Table 9 explores the differential impact of the surcharge during rush hours. Column 1 shows a significant increase in taxi pickups during post-surge rush hour compared to post-surge off-peak hours. This indicates that cabs concentrate their remaining operations into the rush hour period — there are still less pickups overall in the post-surge period but those trips tend to occur during rush hour. In contrast, Column 2 shows no significant change in FHV pickups during rush hour, indicating that the surcharge has not had a comparable effect on the behavior of these services. This divergence between cabs and FHVs is consistent with the idea that cabs are more dependent on rush-hour traffic and more likely to adjust their behavior in response to the surcharge.

6.4. Robustness checks

Finally, we conduct a series of robustness checks to ensure that our results are not driven by the specific choice of boroughs included in the analysis. Table 10 presents the results of these robustness checks, showing that the reduction in collisions holds across different borough subsets. While the significance of the estimates varies slightly, the overall magnitude of the reduction remains consistent, indicating that the policy's impact on traffic safety is robust to different sample selections.

For example, in column 1, we restrict the analysis to Manhattan, which represents over 75% of the taxi zones. The results show a 0.126 (7%) decline in collisions per day per zone, though the estimate is no longer statistically significant. Including Brooklyn in column 2 restores significance, with a decline of 0.133 (7%) collisions per day per zone. Similar results are observed when the Bronx is included in column 3, with 0.116 (7%). When both Brooklyn and the Bronx are included in column 4, we find a negative and significant decline of 0.112 (7%) collisions per day, which falls between the estimates for each borough individually.

Excluding Staten Island, which is geographically distant from Manhattan and has no direct automobile route to the congestion surcharge

area, when considering all other boroughs we observe an even larger decline in collisions of 0.138 (8%) per zone per day. The exclusion of Staten Island is particularly important because it represents a disjoint borough with little direct relevance to the surcharge's impact. By focusing on the areas most affected by the surcharge, we find an even stronger reduction in traffic incidents.

In conclusion, the introduction of the congestion surcharge has led to a significant reduction in traffic activity, collisions, and injuries in the treated areas. These findings are robust across various specifications and suggest that the policy has had a positive impact not only on traffic congestion but also on public safety. By reducing the number of vehicles on the road, the surcharge has contributed to fewer traffic incidents, ultimately making the streets safer for drivers, cyclists, and pedestrians alike.

7. Conclusion

In this article, we find several important effects of the congestion surcharge. First, it appears to work toward its intended effects, it is associated with a significant decline in pickups in the zone the surcharge is applied. This decline is roughly 11% (about 54,000) pickups per day, a decline centered primarily in single rides provided by yellow cabs. FHV programs appear to be quite robust to this surcharge, and even appear to have significant gains in the number of pooled rides they deliver. We find 7 fewer deaths per year in the surcharge region and a significant 465 in total fewer collisions per year, roughly 5% decrease, a nontrivial benefit to the region. Finally, the number of injuries are also less than before the implementation of the surcharge, with a decrease of 9%.

However, there does appear to be some modest spillover effects — collisions may modestly increase north of the surcharge border (but without statistical significance). We find evidence that cabs appear to accumulate their rides more densely during rush hour when the surcharge is imposed relative to when the surcharge is inactive, suggesting there may be some mitigation of congestion-reducing surcharge during critical transportation periods. Finally, careful application of these policies is critical because revenues and therefore the impact of this surcharge is substantial, representing over \$336 million to the city per year (Metropolitan Transportation Authority, 2021).

CRedit authorship contribution statement

Bryan S. Weber: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Moghtaderi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Paolo Cappellari:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Anderson, Michael L., Davis, Lucas W., 2021. Uber and alcohol-related traffic fatalities. Working Paper 29071, In: Working Paper Series, National Bureau of Economic Research.
- Arnott, Richard, Small, Kenneth, 1994. The economics of traffic congestion. Am. Sci. 82.

- Barrios, John M., Hochberg, Yael V., Yi, Hanyi, 2020. The cost of convenience: Ridehailing and traffic fatalities. *J. Oper. Manage.*
- Barron, James, 2018. Where yellow cabs didn't go, green cabs were supposed to thrive. Then came uber..
- Board, Post Editorial, 2022. This 'congestion pricing' plan could kill nyc's economy for good.
- Börjesson, Maria, Eliasson, Jonas, Beser Hugosson, Muriel, Brundell Freij, Karin, 2022. The stockholm congestion charges-5 years on. Effects, acceptability and lessons learnt. *Transp. Policy* 20.
- Börjesson, Maria, Kristoffersson, Ida, 2014. Assessing the welfare effects of congestion charges in a real world setting. *Transp. Res. Part E: Logist. Transp. Rev.* 70, 339–355.
- Börjesson, Maria, Kristoffersson, Ida, 2015. The gothenburg congestion charge. Effects, design and politics. *Transp. Res. Part A: Policy Pr.* 75, 134–146.
- Börjesson, Maria, Kristoffersson, Ida, 2018. The Swedish congestion charges: Ten years on. *Transp. Res. Part A: Policy Pr.* 107, 35–51.
- Buchanan, James M., 1952. The pricing of highway services. *Natl. Tax J.* 5 (2), 97–106.
- Button, Kenneth, 2020. THE transition from Pigou2019s ideas on road pricing to their application. *J. Hist. Econ. Thought* 42 (3), 417–438.
- Cappellari, Paolo, Weber, Bryan S., 2022. An analysis of the new york city traffic volume, vehicle collisions, and safety under COVID-19. *J. Saf. Res.*
- Cetin, Tamer, Deakin, Elizabeth, 2019. Regulation of taxis and the rise of ridesharing. *Transp. Policy* 76, 149–158.
- Cetin, Tamer, Eryigit, Kadir Yasin, 2013. The economic effects of government regulation: Evidence from the new york taxicab market. *Transp. Policy* 25, 169–177.
- Conway, Alison, Tavernier, Nathan, Leal-Tavares, Vanessa, Gharamani, Neda, Chauvet, Lauren, Chiu, Matthew, Bing Yeap, Xiang, 2016. Freight in a bicycle-friendly city: Exploratory analysis with new york city open data. *Transp. Res. Rec.* 2547 (1), 91–101.
- Deakin, Elizabeth, Frick, Karen Trapenberg, Shively, Kevin M., 2010. Markets for dynamic ridesharing?: Case of berkeley, california. *Transp. Res. Rec.* 2187 (1), 131–137.
- Dickerson, Andrew, Peirson, John, Vickerman, Roger, 2000. Road accidents and traffic flows: an econometric investigation. *Economica* 67 (265), 101–121.
- Eliasson, Jonas, Hultkrantz, Lars, Nerhagen, Lena, Smidfelt Rosqvist, Lena, 2009. The stockholm congestion – charging trial 2006: Overview of effects. *Transp. Res. Part A: Policy Pr.* 43, 240–250.
- Else, Peter, 1986. No entry for congestion taxes? *Transp. Res. Part A: Gen.* 20, 99–107.
- Fischer-Baum, Reuben, Bialik, Carl, 2015. Uber is taking millions of manhattan rides away from taxis.
- Fix NYC Advisory Panel, 2018. Advisory panel report.
- Friedman, Milton, 1962. Price Theory: A Provisional Text. Aldine Publishing Company, Chicago, See p. 282 for discussion on entry limits and cab drivers.
- Gold, Josh, Fried, Ben, Miller, Stephen, 2022. Uber: We do support congestion pricing, but with one caveat.
- Green, Colin P., Heywood, John S., Navarro, Maria, 2016. Traffic accidents and the London congestion charge. *J. Public Econ.* 133, 11–22.
- Green, Colin P., Heywood, John S., Paniagua, Maria Navarro, 2020. Did the London congestion charge reduce pollution? *Reg. Sci. Urban Econ.* 84, 103573.
- Hall, Jonathan D., Palsson, Craig, Price, Joseph, 2018. Is uber a substitute or complement for public transit? *J. Urban Econ.* 108, 36–50.
- Han, Lee D., Chin, Shih-Miao, Franzese, Oscar, Hwang, Holing, 2005. Estimating the impact of pickup-and delivery-related illegal parking activities on traffic. *Transp. Res. Rec.* 1906 (1), 49–55.
- Karlström, Anders, Franklin, Joel P., 2009. Behavioral adjustments and equity effects of congestion pricing: Analysis of morning commutes during the stockholm trial. *Transp. Res. Part A: Policy Pr.* 43 (3), 283–296, Stockholm Congestion Charging Trial.
- Kim, Won, Wang, Xiaokun (C.), 2022. Double parking in new york city: a comparison between commercial vehicles and passenger vehicles. *Transportation* 49 (3), 1315–1337.
- Lehe, Lewis, 2019. Downtown congestion pricing in practice. *Transp. Res. Part C: Emerg. Technol.* 100, 200–223.
- Lehe, Lewis, Devunuri, Sairpaneeth, 2021. Large elasticity at introduction. *Res. Transp. Econ.* 101116.
- Lehe, Lewis, Devunuri, Sairpaneeth, Rondon, Javier, Pandey, Ayush, 2021. Taxation of Ride-Hailing. Illinois Center for Transportation/Illinois Department of Transportation.
- Lehe, Lewis, Pandey, Ayush, 2022. Taxi service with heterogeneous drivers and a competitive medallion market. *J. Urban Econ.* 131, 103488.
- Ley, Ana, 2022. Why drivers could soon pay \$23 to reach manhattan.
- Li, Yanchao, Vignon, Daniel, 2024. Do ride-hailing congestion fees in NYC work? *Transp. Res. Part A: Policy Pr.* 190, 104274.
- Liang, Yuan, Yu, Bingjie, Zhang, Xiaojian, Lu, Yi, Yang, Linchuan, 2022. The short-term impact of congestion taxes on ridesourcing demand and traffic congestion: Evidence from chicago. *arXiv preprint arXiv:2207.01793*.
- Markham, Chris, 2022. MTA fare increases announced, effective april 21, 2019.
- Metropolitan Transportation Authority, 2021. MTA 2021 adopted budget 2021–2024. (Accessed April 2023) <https://new.mta.info/document/30186>.
- Moita, Rodrigo, Lucinda, Claudio, Ledo, Bruno, Meyer, Leandro, 2017. The economics of sub-optimal policies for traffic congestion. *J. Transp. Econ. Policy* 51.
- New York City Police Department, 2022. Motor vehicle collision. (Accessed May 2022) <https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95>.
- New York City Taxi and Limousine Commission, 2021. Aggregated reports.
- New York City Taxi and Limousine Commission, 2022. Taxi and limousine commission data trip record data. (Accessed May 2022) <https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.
- New York City Taxi and Limousine Commission, 2023. Reports.
- New York State Department of Taxation and Finance, 2019. New york state department of taxation and finance notice N-19-2. <https://www.tax.ny.gov/pdf/notices/n19-2.pdf>.
- NY State - Division of the Budget, 2018. Governor cuomo announces highlights of the FY 2019 state budget.
- NYC Taxi & Limousine Commission, 2023. New york state's congestion surcharge. <https://www1.nyc.gov/site/tlc/about/congestion-surcharge.page>.
- Olden, Andreas, Møen, Jarle, 2022. The triple difference estimator. *Econom. J.* 25 (3), 531–553.
- OpenData, NYC, 2020. New York City OpenData initiative. (Accessed May 2021) <https://opendata.cityofnewyork.us/>.
- Parry, Ian, 2002. Comparing the efficiency of alternative policies for reducing traffic congestion. *J. Public Econ.* 85, 333–362.
- Peters, Jonathan R., Kramer, Jonathan K., 2012. Just who should pay for what? Vertical equity, transit subsidy and road pricing: The case of new york city. *J. Public Transp.* 15 (2), 6.
- Schillinger, Raymond, 2017. The Death and Life of Helicopter Commuting. Online; (accessed 1 December 2020) <https://www.bloomberg.com/news/articles/2017-08-10/the-return-of-rooftop-helicopter-commuting-in-new-york-city>.
- Shefer, Daniel, Rietveld, Piet, 1997. Congestion and safety on highways: Towards an analytical model. *Urban Stud.* 34 (4), 679–692.
- Singichetti, Bhavna, Conklin, Jamie L, Hassmiller Lich, Kristen, Sabounchi, Nasim S, Naumann, Rebecca B, 2021. Congestion pricing policies and safety implications: a scoping review. *J. Urban Heal.* 1–18.
- Spectrum News, 2018. Yellow cab surcharge is coming in january.
- Sugiyama, Naoki, Nagatani, Takashi, 2012. Multiple-vehicle collision induced by a sudden stop in traffic flow. *Phys. Lett. A* 376 (22), 1803–1806.
- Tang, Tie-Qiao, Li, Yan, Huang, Hai-Jun, 2009. The effects of taxi ON traffic flow. *Internat. J. Modern Phys. C* 20 (10), 1537–1546.
- Vickrey, William, 1968. Automobile accidents, tort law, externalities, and insurance: An economist's critique. *Law Contemp. Probl.* 33 (3), 464–487.
- Vitaliano, Donald F., Held, James, 1991. Road accident external effects: an empirical assessment. *Appl. Econ.* 23 (2), 373–378.
- Wallsten, Scott, 2015. The competitive effects of the sharing economy: how is uber changing taxis. *Technol. Policy Inst.* 22, 1–21.
- West, Jens, Börjesson, Maria, 2020. The gothenburg congestion charges: cost–benefit analysis and distribution effects. *Transportation* 47 (1), 145–174.
- Yuan, Yanhong, Li, Guomin, Zhang, Rongxia, Li, Wei, Fan, Qingqing, 2020. Congestion charges in mega cities: On affection and effectiveness. *J. Environ. Account. Manag.* 8 (1), 41–54.